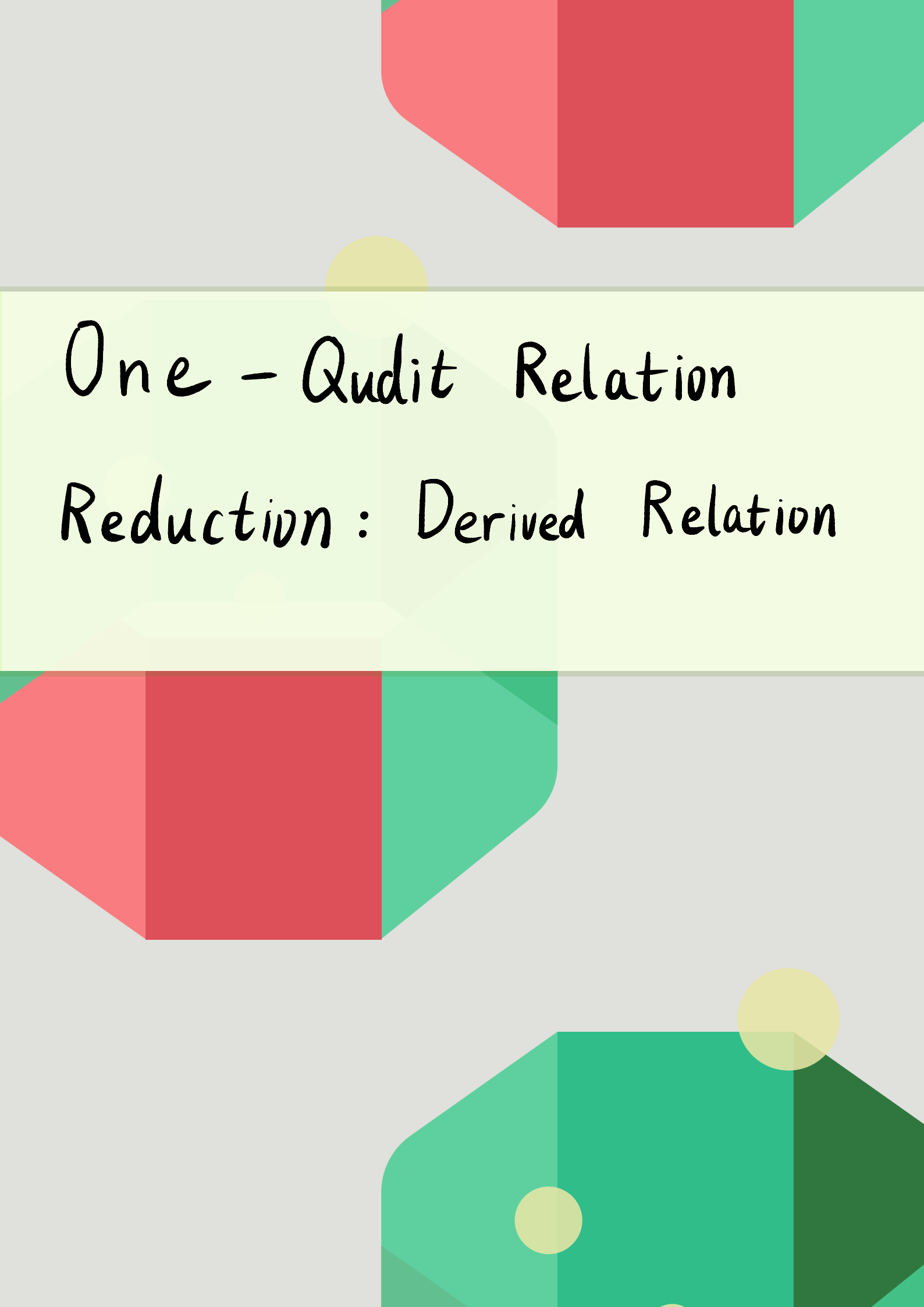




One - Qudit Relation

Reduction : Derived Relation

Def: $M_a |x\rangle = |ax\rangle \quad a \in \mathbb{Z}_p^* \quad \text{---} \boxed{M_a} \text{---} = \text{---} \boxed{H} \text{---} \boxed{S^a} \text{---} \boxed{H} \text{---} \boxed{S^{a^{-1}}} \text{---} \boxed{H} \text{---} \boxed{S^a} \text{---}$

Lem 1 $M_a^{-1} = M_{a^{-1}}$, a^{-1} is the inverse of a modulo p

Proof: For all $x \in \mathbb{Z}_p$, $M_{a^{-1}} M_a |x\rangle = M_{a^{-1}} |ax\rangle = |a^{-1}ax\rangle = |x\rangle$ and $M_a M_{a^{-1}} |x\rangle = |x\rangle$.

Hence, $M_{a^{-1}} M_a = M_a M_{a^{-1}} = I$. It follows that $M_{a^{-1}} = M_a^{-1}$. □

Corollary 2 $M_a^\dagger = M_{a^{-1}}$.

Proof: Since M_a is unitary, $M_a^\dagger = M_a^{-1}$. By Lem 1, $M_a^\dagger = M_{a^{-1}}$. □

Lem 3 $\frac{x}{z} \boxed{M_a} \frac{x^{a^{-1}}}{z^a} \quad \frac{x^a}{z^{a^{-1}}} \boxed{M_a} \frac{x}{z} \quad M_a = \frac{x}{z} \begin{bmatrix} x & z \\ a^{-1} & 0 \\ 0 & a \end{bmatrix}$

Proof: $\frac{x}{z} \boxed{H} \frac{z}{x^{-1}} \boxed{S^a} \frac{z}{x^{-1}z^a} \boxed{H} \frac{x^{-1}}{x^a z^{-1}} \boxed{S^{a^{-1}}} \frac{x^{a^{-1}}}{x^a} \boxed{H} \frac{x^{a^{-1}} z^{-1}}{z^a} \boxed{S^a} \frac{x^{a^{-1}} z z^{-1}}{z^a} = \frac{x^{a^{-1}}}{z^a}$
 $\frac{x^a}{z^{a^{-1}}} \boxed{H} \frac{z^a}{x^{a^{-1}}} \boxed{S^a} \frac{z^a}{x^{-a}} \boxed{H} \frac{x^{-a}}{x^a z^{-1}} \boxed{S^{a^{-1}}} \frac{x^a z^{-1}}{x} \boxed{H} \frac{x z^{-a}}{z} \boxed{S^a} \frac{x}{z} = \frac{x}{z}$ ✓
 $\frac{x^a}{z^{a^{-1}}} \boxed{H} \frac{z^a}{x^{a^{-1}}} \boxed{S^a} \frac{z^a}{x^{-a}} \boxed{H} \frac{x^{-a}}{x^a z^{-1}} \boxed{S^{a^{-1}}} \frac{x^a z^{-1}}{x} \boxed{H} \frac{x z^{-a}}{z} \boxed{S^a} \frac{x}{z} = \frac{x}{z}$ ✓

$$S^a = \frac{x}{z} \begin{bmatrix} x & z \\ a^{-1} & 0 \\ 0 & a \end{bmatrix}$$

$$x \boxed{S^a} = x z^a \quad z \boxed{S^a} = z$$

Lem 4 $\text{---} \boxed{H} \text{---} \boxed{M_a} \text{---} \cong \text{---} \boxed{M_{a^{-1}}} \text{---} \boxed{H} \text{---} \quad \text{---} \boxed{M_a} \text{---} \boxed{H} \text{---} \cong \text{---} \boxed{H} \text{---} \boxed{M_{a^{-1}}} \text{---}$

Proof: $\frac{x}{z} \boxed{H} \frac{z}{x^{-1}} \boxed{M_a} \frac{z^a}{x^{a^{-1}}} \cong \frac{x}{z} \boxed{M_{a^{-1}}} \frac{x^a}{z^{a^{-1}}} \boxed{H} \frac{z^a}{x^{a^{-1}}} \quad \checkmark$

$\frac{x}{z} \boxed{M_a} \frac{x^{a^{-1}}}{z^a} \boxed{H} \frac{z^{a^{-1}}}{x^{-a}} \cong \frac{x}{z} \boxed{H} \frac{z}{x^{-1}} \boxed{M_{a^{-1}}} \frac{z^{a^{-1}}}{x^{-a}} \quad \checkmark$

Lem 5 $\text{---} \boxed{S} \text{---} \boxed{M_a} \text{---} \cong \text{---} \boxed{M_a} \text{---} \boxed{S^{a^2}} \text{---} \quad \text{---} \boxed{M_a} \text{---} \boxed{S} \text{---} \cong \text{---} \boxed{S^{a^2}} \text{---} \boxed{M_a} \text{---}$

Proof: $\frac{x}{z} \boxed{S} \frac{x z}{z} \boxed{M_a} \frac{x^{a^{-1}} z^a}{z^a} \cong \frac{x}{z} \boxed{M_a} \frac{x^{a^{-1}}}{z^a} \boxed{S^{a^2}} \frac{x^{a^{-1}} z^a}{z^a} \quad \checkmark$

$\frac{x}{z} \boxed{M_a} \frac{x^{a^{-1}}}{z^a} \boxed{S} \frac{x^{a^{-1}} z^{a^{-1}}}{z^a} \cong \frac{x}{z} \boxed{S^{a^2}} \frac{x z^{a^2}}{z} \boxed{M_a} \frac{x^{a^{-1}} z^{a^{-1}}}{z^a} \quad \checkmark$

Lem 6 $\underbrace{\begin{array}{c} \boxed{M_a} \cdots \boxed{M_a} \\ k \quad k \in \mathbb{Z}_p \end{array}} \hat{=} \boxed{M_{a^k}}$

$$\frac{x}{z} \boxed{M_a} \frac{x^{a^{-1}}}{z^a} \boxed{M_a} \frac{(x^{a^{-1}})^{a^{-1}}}{z^{a^2}} = \frac{x}{z} \boxed{M_{a^2}} \frac{x^{(a^2)^{-1}}}{z^{a^2}} = x^{a^{-2}}$$

Proof: Base Case: $k=1$ ✓

Induction Hypothesis: The statement holds for $k \geq 1$.

$$\frac{x}{z} \underbrace{\boxed{M_a} \frac{x^{a^{-1}}}{z^a} \cdots \boxed{M_a} \frac{x^{a^{-k}}}{z^{a^k}}}_k \hat{=} \frac{x}{z} \boxed{M_{a^k}} \frac{x^{(a^k)^{-1}}}{z^{a^k}} = x^{a^{-k}}$$

Induction Step: The statement holds for $k+1$. Note that $\forall a, k \in \mathbb{Z}_p^*, (a^k)^{-1} = a^{-k}$ since $a^{-k}a^k = 1$.

$$\frac{x}{z} \underbrace{\boxed{M_a} \frac{x^{a^{-1}}}{z^a} \cdots \boxed{M_a} \frac{x^{a^{-k}}}{z^{a^k}}}_k \boxed{M_a} \frac{(x^{a^{-1}})^{a^{-k}}}{(z^a)^{a^k}} = \frac{x}{z} \underbrace{\boxed{M_a} \frac{x^{a^{-1}}}{z^a} \cdots \boxed{M_a} \frac{x^{a^{-k}}}{z^{a^k}}}_{k+1} \hat{=} \frac{x}{z} \boxed{M_{a^{k+1}}} \frac{x^{(a^{k+1})^{-1}}}{z^{a^{k+1}}} = x^{a^{-(k+1)}}$$

□

Lem 7 $\boxed{H^2} \hat{=} \boxed{M_{-1}}$

Proof: $\frac{x}{z} \boxed{H^2} \frac{x^{-1}}{z^{-1}} \hat{=} \frac{x}{z} \boxed{M_{-1}} \frac{x^{-1}}{z^{-1}}$

□

Corollary 8 $\boxed{M_{(-1)^2}} \hat{=} \boxed{M_{a^p}} = I \quad a \in \mathbb{Z}_p^*$

Proof: $I = \boxed{H^4} \xrightarrow{\text{Lem 11}} \boxed{M_{-1}} \boxed{M_{-1}} \xrightarrow{\text{Lem 10}} \boxed{M_{(-1)^2}} = \boxed{M_1} = \boxed{M_{a^p}}$

Lem 5 $\boxed{M_a} \boxed{S} \hat{=} \boxed{S^{a^2}} \boxed{M_a}$ Lem 7 $\boxed{H^2} \hat{=} \boxed{M_{-1}}$

Corollary 9 $\boxed{H^2} \boxed{S} \hat{=} \boxed{S} \boxed{H^2} \quad (-1)^{-2} = ((-1)^2)^{-1} = 1$

Proof: LHS = $\boxed{H^2} \boxed{S} \xrightarrow{\text{Lem 7}} \boxed{M_{-1}} \boxed{S} \xrightarrow{\text{Lem 5}} \boxed{S} \boxed{M_{-1}} \xrightarrow{\text{Lem 7}} \boxed{S} \boxed{H^2} = \text{RHS}$

$$D_1: -[M_a] - = -[H] - [S^a] - [H] - [S^{a^{-1}}] - [H] - [S^a] - \quad a \in \mathbb{Z}_p^*$$

$$\text{Corollary 9} \quad -[H^2] - [S] - \triangleq -[S] - [H^2] -$$

$$\text{Corollary 10} \quad H^4 \triangleq I$$

$$R_4: -[H] - [M_g] - = -[M_{g^{-1}}] - [H] - \quad R_4^1: -[M_g] - [H] - = -[H] - [M_{g^{-1}}] -$$

$$\text{Corollary 9}^* \quad R_3^3: -[H^2] - [M_g] - = -[M_g] - [H^2] -$$

$$\text{Proof: LHS} = -[H^2] - [M_g] - \stackrel{D_1}{=} -[H^2] - [H] - [S^g] - [H] - [S^{g^{-1}}] - [H] - [S^g] -$$

$$\stackrel{R_3^2}{=} -[H] - [S^g] - [H] - [S^{g^{-1}}] - [H] - [S^g] - [H^2] - \stackrel{D_1}{=} -[M_g] - [H^2] - = \text{RHS} \quad \square$$

$$\text{Corollary 9}^{**} \quad R_4^2: -[H^3] - [M_g] - = -[M_{g^{-1}}] - [H^3] -$$

$$R_4^3: -[M_g] - [H^3] - = -[H^3] - [M_{g^{-1}}] -$$

$$\text{Proof: } R_4^2 \cdot \text{LHS} = H^3 M_g = H [H^2 M_g] \stackrel{R_3^3}{=} H [M_g] H^2 \stackrel{R_4^1}{=} M_{g^{-1}} H H^2 = M_{g^{-1}} H^3 = R_4^2 \cdot \text{RHS}.$$

$$R_4^3 \cdot \text{LHS} = M_g H^3 = [M_g H^2] H \stackrel{R_3^3}{=} H^2 [M_g H] \stackrel{R_4^1}{=} H^2 H M_{g^{-1}} = H^3 M_{g^{-1}} = R_4^3 \cdot \text{RHS} \quad \square$$

Corollary 10 $H^4 \triangleq I$.

Proof: $H^4 \xrightarrow[\text{Lem 7}]{=} \boxed{M_{-1}} \boxed{M_{-1}} \xrightarrow[\text{Lem 6}]{=} \boxed{M_{(-1)^2}} \xrightarrow[\text{Cor 8}]{=} I = \text{RHS.}$

□

Lem 11 D_2 : $\boxed{A_{ab}} = \boxed{M_{-a}} \boxed{H^2} \boxed{S^{-ab}} \boxed{H} \quad a > 0, b \in \mathbb{Z}_p$

Proof: Recall that $\boxed{A_{ab}} = \boxed{H^{-1}} \boxed{S^{-a}} \boxed{H^{-1}} \boxed{S^{-1/a}} \boxed{H^{-1}} \boxed{S^{-a(b+1)}} \boxed{H} \quad \checkmark$

$S^{-a} S^{-ab} = S^{-a-ab} = S^{-a(b+1)}$

$A_{ab} \xrightarrow[\text{Cor 10}]{=} \boxed{H^3 S^{-a}} H^3 S^{-1/a} H^3 S^{-a} S^{-ab} H$

$\xrightarrow[\text{Cor 9}]{=} H S^{-a} \boxed{H^2 H^3} S^{-1/a} H^3 S^{-a} S^{-ab} H$

$\xrightarrow[\text{Cor 10}]{=} H S^{-a} H S^{-1/a} \boxed{H^3 S^{-a}} S^{-ab} H$

$\xrightarrow[\text{Cor 9}]{=} \boxed{H S^{-a} H S^{-1/a} H S^{-a}} H^2 S^{-ab} H$

$\xrightarrow[\text{D}_1]{=} M_{-a} H^2 S^{-ab} H$

□

Lem 12 D_3 : $\boxed{A_{ob}} = \boxed{M_{b^{-1}}} \boxed{S^{-1/b}} \quad b \in \mathbb{Z}_p^*$

Proof: Recall that $\boxed{A_{ob}} = \boxed{H^{-1}} \boxed{S^{1/b}} \boxed{H^{-1}} \boxed{S^b} \boxed{H} \quad \checkmark$

$$A_{ob} = H^{-1} S^{1/b} H^{-1} S^b H \stackrel{\text{Cor 10}}{\cong} \boxed{H^3 S^{1/b}} H^3 S^b H \stackrel{\text{Cor 9}}{\cong} H S^{1/b} \boxed{H^2 H^3} S^b H$$

$$\stackrel{\text{Cor 10}}{\cong} H S^{1/b} H S^b H = \boxed{H S^{1/b} H S^b H S^{1/b}} S^{-1/b} \stackrel{D_1}{\cong} M_{b^{-1}} S^{-1/b}$$

□

Lem 12* R_5 : $(M_g)^k = M_{g^k}$ implies $M_a \cdot M_b = M_{ab}$, where $a, b, k \in \mathbb{Z}_p$

Proof: Suppose $a = g^{k_1}$ $b = g^{k_2}$ for $k_1, k_2 \in \mathbb{Z}_p$. Then $ab = g^{k_1} g^{k_2} = g^{k_1+k_2}$

Hence $RHS = M_{ab} = M_{g^{k_1+k_2}} \stackrel{R_5}{=} (M_g)^{k_1+k_2}$

$LHS = M_a M_b = M_{g^{k_1}} M_{g^{k_2}} \stackrel{R_5}{=} (M_g)^{k_1} (M_g)^{k_2} = (M_g)^{k_1+k_2} = RHS.$

□

Lem 12** D'_3 : $\boxed{A_{ob}} = \boxed{H} \boxed{S^{1/b}} \boxed{H} \boxed{S^b} \boxed{H} \quad b \in \mathbb{Z}_p^*$

Proof: Recall that $\boxed{A_{ob}} = \boxed{H^{-1}} \boxed{S^{1/b}} \boxed{H^{-1}} \boxed{S^b} \boxed{H} \quad \checkmark$

$$A_{ob} = H^{-1} S^{1/b} H^{-1} S^b H \stackrel{\text{Cor 10}}{\cong} \boxed{H^3 S^{1/b}} H^3 S^b H \stackrel{\text{Cor 9}}{\cong} H S^{1/b} \boxed{H^2 H^3} S^b H$$

$$\stackrel{\text{Cor 10}}{\cong} H S^{1/b} H S^b H$$

□

Lem 12*** D_3^2 : $\boxed{A_{ob}} = \boxed{H^3} \boxed{S^{1/b}} \boxed{H} \boxed{S^b} \boxed{H^3}$

Proof: $RHS = \boxed{H^3 S^{1/b} H S^b} H^3 \stackrel{R_5^2}{=} H S^{1/b} H S^b \boxed{H^2 H^3} \stackrel{R_5^1}{=} H S^{1/b} H S^b H \stackrel{D_3^1}{=} A_{ob} = LHS.$

□